

**Rollno:241001056**

## **SOLVE A GRAPH USING DEPTH FIRST SEARCH(DFS)**

**Python code:**

```
def dfs(graph, start, visited=None):
    if visited is None:
        visited = set()

    visited.add(start)
    print(start, end=" ")

    for neighbor in graph[start]:
        if neighbor not in visited:
            dfs(graph, neighbor, visited)

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
```

```
'D': [],  
'E': ['F'],  
'F': []  
}
```

```
print("DFS traversal from node A:")  
dfs(graph, 'A')
```

**output:**

DFS traversal from node A:

A B D E F C

**Result:**

Thus the program of implementing DFS traversal on a graph to explore all nodes was executed successfully.